

3. Project Design Phase

Architecture, data flow, and configuration design

System Architecture

The solution has four main layers: spreadsheet source, Import Set staging table, Transform Map processing, and Employee Test target table. Reporting and dashboard components read from the target table after data validation.

Data Flow

Excel spreadsheet → Load Data → Employee Import staging table → Sample Spreadsheet Import Transform Map → Employee Test target table → validation → reports and dashboard.

Database / Table Design

Target table: Employee Test (u_employee_test). Fields: Employee ID (String), Employee Name (String), Email (String), Department (String), and Location (String). The staging table is Employee Import (u_employee_import), populated from the spreadsheet.

Transform Map Design

Name: Sample Spreadsheet Import. Source Table: Employee Import. Target Table: Employee Test. Use Auto Map Matching Fields where applicable, then Mapping Assist to confirm source-to-target mappings. Save the map before executing Transform.

Field Mapping

- Employee ID → Employee ID
- Employee Name → Employee Name
- Email → Email
- Department → Department
- Location → Location

Coalesce Design

Set the Employee ID field map's Coalesce property to True. This makes Employee ID the match key for deciding whether an incoming row should update an existing target record or insert a new one. Confirm the setting on the field map and save.

Reporting Design

- Employees by Department: Pie chart, count aggregation, group by Department.
- Employees by Location: Bar chart, count aggregation, group by Location.
- Employee List Report: List report showing Employee ID, Employee Name, Email, Department, and Location.
- Dashboard: Employee Analytics Dashboards, containing all three reports.

Validation Design

After transformation, inspect the Employee Test list and Transform History. Compare row counts and sample field values against the spreadsheet. For the repeat-import test, verify that changed matching IDs update, new IDs insert, and an identical reimport does not create additional records.